

Latent-Space Steering of Motion Diffusion Models for Modular Humanoid Locomotion

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Abstract—Humanoid locomotion requires both diverse whole-body motion generation and physically stable control. Recent modular frameworks combine motion diffusion models with reinforcement-learning-based tracking policies, but task adaptation is often still tied to the conditioning interface of the diffusion planner. This project investigates whether Diffusion Steering via Reinforcement Learning (DSRL) can make this hierarchy more modular. We keep a pretrained motion diffusion planner fixed, learn a lightweight latent steering policy to adapt the planner to downstream rewards, and use a robust RL-based tracking policy to execute the generated kinematic targets on the G1 humanoid robot.

I. MOTIVATION

Humanoid locomotion requires both diverse whole-body motion generation and physically stable control. A humanoid robot must adapt its gait, posture, and contact sequence to different downstream tasks, such as navigation, obstacle traversal, and object interaction. Training a single end-to-end policy for all of these behaviors is difficult because motion generation, task planning, and low-level physical control become entangled in one model.

A promising alternative is a modular framework. In this view, a motion generation module proposes high-level kinematic motions, while a tracking policy executes those motions under physics. Recent work such as CLoSD demonstrates the strength of this approach by connecting a real-time motion diffusion planner with a physics-based tracking controller in a closed loop [1]. The planner generates short-horizon motion targets from the current motion context and task conditioning, the tracking policy executes them, and the realized motion is fed back to the planner for the next planning step. This structure allows the system to combine the diversity of motion diffusion models with the physical feasibility of reinforcement-learning-based tracking controllers.

However, the modularity of existing frameworks is still limited. In CLoSD, task adaptation is mainly handled through the diffusion planner’s conditioning interface, such as text prompts or target joint locations [1]. This works when the downstream task can be naturally expressed through that interface. But many humanoid locomotion tasks require more complex task feedback or local motion-selection strategies. For example, obstacle traversal may require preparing the body before reaching the obstacle, changing gait, crouching, jumping, or selecting an appropriate contact sequence. These behaviors are not always well captured by a fixed prompt or target-location

interface. When the task representation changes, the diffusion planner may need to be redesigned, reconditioned, or retrained.

This project investigates whether Diffusion Steering via Reinforcement Learning (DSRL) can make this framework more modular. DSRL keeps the pretrained diffusion model fixed and learns a lightweight policy over the latent noise space of the diffusion process [2]. Instead of updating the diffusion model parameters, the steering policy selects the initial noise input so that the frozen diffusion model produces outputs that better satisfy downstream rewards.

Our key idea is to transfer DSRL from action generation to motion planning. The motion diffusion model serves as a reusable motion prior. The low-level tracking policy serves as a reusable physical executor. Task adaptation is handled outside the diffusion model by a latent steering policy trained with reinforcement learning. The goal of this project is to test whether useful and physically trackable motion targets for downstream humanoid locomotion tasks can be obtained by steering the initial latent noise of a frozen motion diffusion planner, without retraining the planner itself.

II. METHOD 1: DIFFUSION POLICY OF DSRL

Diffusion policies have become a powerful generative model for robotics because they can represent multimodal behavior distributions. Unlike standard behavior cloning, which may average multiple valid actions into an invalid action, a diffusion policy can model diverse action or trajectory candidates. However, a pretrained diffusion policy is still constrained by the behavior distribution of its training data. When the downstream objective changes, directly fine-tuning the full diffusion model can be computationally expensive and unstable.

DSRL addresses this problem by adapting a frozen diffusion model through latent-space reinforcement learning [2]. A steering policy selects the initial noise vector of the diffusion process. The frozen diffusion model then maps this steered noise into an output trajectory. Since the diffusion model weights are not updated, the pretrained behavioral prior is preserved, while task adaptation is achieved through the steering policy.

In our framework, the diffusion model does not directly output motor actions. Following the CLoSD-style hierarchy, it generates a short-horizon kinematic motion target for a low-level tracking controller [1]. The original diffusion planner samples its initial latent from Gaussian noise. We replace this random initialization with a learned steering policy. Given the

current robot state and task information, the steering policy outputs or modifies the initial latent noise. The frozen motion diffusion planner then denoises this latent into a future motion target.

This design separates task adaptation from motion-prior learning. The motion diffusion model is trained once and reused as a general motion prior. For a new downstream locomotion task, we do not retrain the diffusion planner. Instead, we train only the lightweight steering policy using task rewards. The steering policy learns which region of the diffusion latent space is likely to produce motion targets that help the robot solve the task.

Formally, the overall planning process can be described as follows. At each high-level planning step, the system observes the current motion state and task context. The steering policy selects an initial latent noise vector. The frozen diffusion planner generates a kinematic target trajectory from this latent. The tracking policy then executes the target in physics. The resulting state is fed back to the planner, closing the loop between latent steering, motion generation, and physical execution.

The central hypothesis of Method 1 is that downstream humanoid locomotion tasks can be solved by steering the initial latent noise of a frozen motion diffusion planner, while preserving the planner as a reusable motion prior.

III. METHOD 2: RL-BASED TRACKING POLICY

The high-level diffusion planner generates kinematic target trajectories, but these targets are not guaranteed to be physically feasible for the G1 humanoid robot. This issue becomes more important when the high-level planner is optimized with task rewards through latent steering. To maximize rewards, the steering policy may push the diffusion planner toward aggressive or out-of-distribution motions, such as high jumps, fast transitions, or unstable postures. Therefore, the low-level controller must not simply imitate the target motion. It must robustly track useful targets while maintaining balance and respecting the robot’s physical constraints.

Previous humanoid tracking methods often use reinforcement learning with manually designed tracking rewards. For example, DeepMimic-style controllers minimize errors between the reference motion and the simulated robot state at each timestep [3]. This approach can produce high-quality tracking, but it requires careful reward design and weight tuning. For a complex humanoid such as G1, which has many degrees of freedom, manually balancing joint tracking, body tracking, velocity tracking, balance, and smoothness terms is inefficient and difficult to scale.

Other approaches reduce manual reward engineering by introducing learned style or motion priors, such as adversarial discriminators [4]. However, many of these methods still impose strict reference matching. This can become problematic in our setting because the high-level diffusion planner may intentionally generate motions that deviate from the original motion dataset to satisfy downstream task rewards. A tracker

that is too rigid may fail when the target trajectory is dynamically aggressive or slightly out of distribution.

To address this, we use a robust RL-based tracking policy trained on large-scale retargeted motion data. The role of the tracker is not only to follow the generated target, but also to recover from imperfect or physically unstable targets. Recent frameworks such as PHC show that a single large-scale humanoid controller can track diverse motion commands by training on broad human motion datasets and using motion history to infer dynamic context [5]. CLoSD also relies on this type of robust low-level tracking policy [1].

For deployment on the G1 robot, however, human-avatar motion data cannot be used directly. The morphology and kinematic structure of G1 differ from those of standard human motion-capture skeletons. Therefore, kinematic retargeting is necessary before training the low-level tracker. In this project, we adopt a GEAR-SONIC-style tracking pipeline, which combines kinematic retargeting with large-scale domain randomization [6]. Retargeting aligns the motion dataset with the G1 morphology, while domain randomization improves robustness to dynamics mismatch and sim-to-real gaps.

The final tracking module receives the kinematic target trajectory generated by the steered diffusion planner and outputs low-level joint actions for the G1 robot. Its objective is to execute the target as faithfully as possible while preserving balance, smoothness, and physical feasibility. In the overall system, Method 1 provides task-adaptive motion targets through latent steering, and Method 2 provides a robust physical executor that converts those targets into stable humanoid behavior.

Together, these two components form a modular humanoid locomotion framework: the frozen motion diffusion model provides a reusable motion prior, the DSRL steering policy provides task-specific adaptation, and the RL-based tracking policy provides physically feasible execution on the target robot.

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